

Negatively Strain-Dependent Electrical Resistance of Magnetically Arranged Nickel Composites: Application to Highly Stretchable Electrodes and Stretchable Lighting Devices

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In recent years, there has been an increasing interest in stretchable electronics. It has great potential in future electronics such as flexible displays, wearable electronics and biomedical applications.^[1,2] With recent progress in stretchable electronics, electronic devices on stretchable substrates can be protected from mechanical deformation by allowing the stretchable electrodes to undergo most of the mechanical stress.^[3–6] Previous stretchable electrodes were mainly realized by either vacuum deposition or inkjet-printing of metal thin films on elastomeric substrates.^[7–12] These stretchable electrodes show outstanding electrical performance at relatively low tensile strain conditions typically under 30% elongation.

In order to realize stretchable electrodes that can well sustain electrical performance at higher tensile strain conditions (over 30% elongation), conductive composite materials have also been widely used. Generally, conductive composite materials which are composed of elastomeric matrix and conductive fillers such as metal powder,^[13,14] carbon black^[15,16] and graphite,^[17] show relatively high initial resistance to be used as interconnection between electronic devices. Recently carbon nanotube (CNT)-based conductive composite materials have been reported for stretchable electrode applications.^[1,18,19] With special treatment, these materials show superior electrical property at over 30% tensile strain. In view of practical use, however, they require complex fabrication processes to attain high initial conductivity and show significant decrease in conductivity with increasing tensile strain. Furthermore, patterning of the embedded conductive materials is also challenging for previously reported methods. In the Supporting Information, SI, Table S1 summarizes recent representative conductive composites and their electrical characteristics with mechanical stretching, including our results. This comparative chart shows a typical tendency of most conductive composites. It is hard to maintain the initial conductivity under the tensile strain even if

they have remarkable initial conductivity and can mechanically sustain at high tensile strain.

It is noted that most conductive materials show positively strain-dependent resistance change, i.e., resistance increases with applied tensile strain, while our material shows opposite behavior. This unique property can be used as a device supplying power at high tensile strain conditions because the electrodes show low resistance when elongated. In addition, when combined with the inorganic metal films with very low initial resistance, high-performance stretchable electrodes with low resistance at both low and high tensile strain conditions can be implemented. Furthermore, our materials are ferromagnetic and thus can be patterned by a patterned magnetic field in a simple manner. Therefore, depending on the configuration of our composite-only and metal film electrodes, we can fabricate a new conceptual device, such as a stretchable lighting or display device with constant image resolution even under tensile strain conditions.

In this study, we report a novel property of negatively strain-dependent electrical resistance change of a composite where nickel powder conductive filler is embedded in an elastomeric medium. When the conductive fillers in the composite were arranged in a direction perpendicular to the strain plane under the magnetic field, the composite showed negative strain-dependency. When we used the patterned magnetic field with the pre-patterned ferromagnetic iron structure, patterned and embedded composite electrodes were also simultaneously formed. Details of this process will be later described. In order to fabricate a highly stretchable electrode with a very low initial resistance (a few ohms for 10 mm long electrode), we combined the inkjet-printed silver thin film on the magnetically patterned and arranged conductive composite materials. Silver thin film provides very low initial resistance and maintains the value at low tensile strain range. As the tensile strain increases, the conductive composite with the negative strain-dependency takes over and maintains its resistance low at high tensile strain range. In addition, we also implemented a resolution-sustainable, stretchable lighting device, by appropriately configuring both composite-only and silver-covered composite electrodes as interconnecting lines for surface-mount light-emitting diodes (LEDs). In this conceptual device, hidden LED pixels are located between on-pixels and connected to the composite-only electrodes so that they supply power to and turn on the hidden pixels, maintaining the resolution of the lighting

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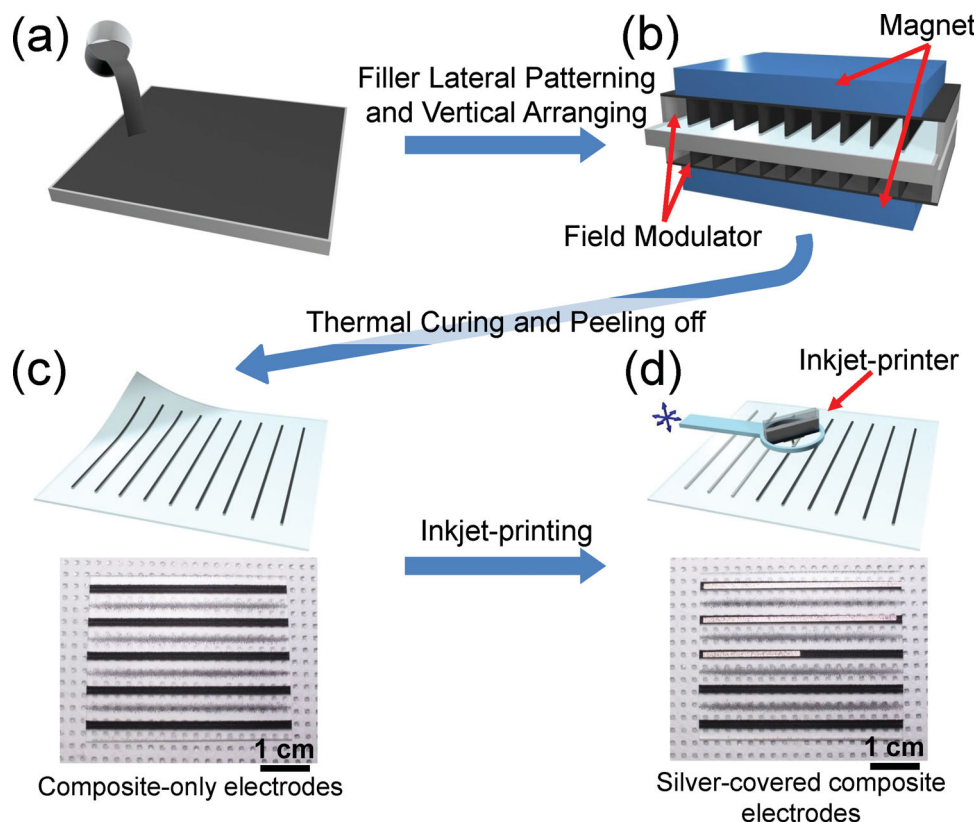


Figure 1. Fabrication process of stretchable electrodes. (a) Nickel/PDMS mixture casting onto aluminum mold. (b) Nickel filler arranging and patterning by the magnetic field modulator. (c) Cured exposed composite-only electrodes. (d) Inkjet-printing of silver electrodes onto exposed composite-only electrodes.

device, when the distance between on-pixels increases under the tensile strain.

To implement the stretchable electrodes, we used poly(dimethylsiloxane) (PDMS; Sylgard 184 from Dow Corning) for elastomeric medium and nickel powder (203904; average particle size $\sim 5\ \mu\text{m}$, 99.99% from Sigma Aldrich) for conductive fillers. Filler volume ratio can be varied to optimize the performance of the conductive composite. **Figure 1** shows fabrication flow of the composite-only and silver-covered composite electrodes. PDMS/nickel mixture was cast onto a flat aluminum mold (see Figure 1a) and then the mold was placed between two pre-made iron structures (we will call the structure “magnetic field modulator”) and two neodymium permanent magnets as shown in Figure 1b. During this process, magnetic field from magnets is focused at closest confronting area of the two magnetic field modulators. The nickel powder inside the mixture moves laterally under the focused area, making electrode patterns similar to the shape of the magnetic field modulator. At the same time, the nickel powder was vertically arranged due to the magnetic field, direction of which is perpendicular to the composite plane. It is noted that, we can produce various shapes such as wavy lines or circles in addition to the straight lines if we use magnetic field modulators with corresponding patterns. However, our method has limitation in producing very fine features because feature sizes are directly related to the minimum feature size of the shape of the magnetic field modulator and the thickness of the nickel-embedded

PDMS. Since our field modulator is made by an electrical discharge machining (EDM) method and has high aspect ratio structure, its pattern pitch (or line width and spacing) is limited to around $100\ \mu\text{m}$ due to the intrinsic EDM limitation. The important point is that the designed pattern pitch and PDMS thickness should be similar in order to obtain desirable pattern quality. The time required for patterning/arrangement process depends on volume ratio of the nickel powder, mixture viscosity, and magnetic field strength at the confronting surface. Finally, the patterned and arranged mixture was thermally cured. The fully cured arranged conductive composite electrodes (we call this “exposed composite-only electrodes”) have 10-mm length, 1-mm width and 1-mm thickness as shown in Figure 1c. On top of the fully cured exposed composite-only electrodes, silver thin films were inkjet-printed (see Figure 1d, we call this “exposed silver-covered composite electrodes”) by using the printing conditions, details of which have been reported in our previous papers.^[11,12] For comparison, we also fabricated two additional types of electrodes: nickel conductive composite electrodes without exposure to magnetic field (we call this “unexposed composite-only electrode”) and inkjet-printed silver electrodes on the PDMS substrate.

Figure 2a shows resistance change of the unexposed and exposed composite-only electrodes with the tensile strain for different filler concentrations of 5.3 to 18.3 vol% (lines with symbol). It is noted that volume fraction was calculated from the measured weight of PDMS and nickel powder and literature

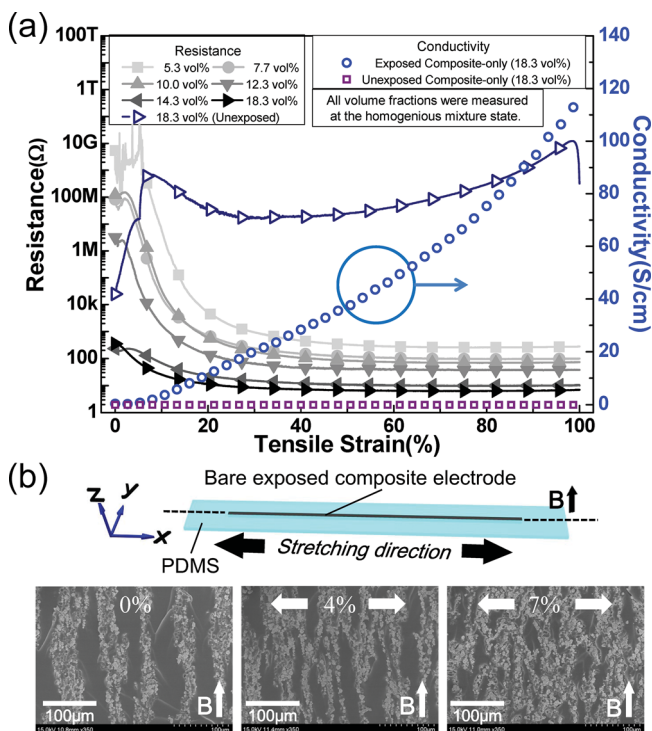


Figure 2. (a) Resistance and conductivity change in terms of tensile strain for exposed and unexposed composite-only electrodes. (All volume fractions were measured at the homogeneous mixture state.) (b) Schematic illustration of the exposed composite-only electrode and its cross sectional SEM images under various tensile strain. (Magnetic field direction: z , stretching direction: x , SEM image: xz -plane through dashed line)

values of their density (1.03 and 8.908 g/cm³ for PDMS and nickel, respectively) and indicates the volume fraction at the homogeneous mixture state. For exposed composite, resistance at the same tensile stress decreases with the filler concentration because the more nickel powder exists per volume, the more conduction paths are formed in the composite electrodes. The same behavior was also observed for the unexposed ones, which is not shown here. Resistance of the exposed composite-only electrodes rapidly decreases with the tensile strain at low strain region, and then is saturated at around 40% strain. Electrical conductivity that was calculated from resistance and volume of the composite was about 115 S/cm at 100% tensile strain. However, for unexposed composite, opposite properties, i.e. increase of resistance with the tensile stress, have been observed. In addition, its initial resistance was also very large in comparison with the exposed composite.

As mentioned before, it should be noted that the conductive composite materials that have been previously reported generally showed properties of increasing resistance with the tensile strain like our unexposed composite.^[13–19] The exposed composite-only electrodes showed conductive path increasing, which is verified from the cross-sectional scanning electron microscope (SEM) image with the tensile strain. Figure 2b shows that electrical conduction paths are more formed between nickel powders as the tensile strain increases. This unique property of the exposed composite-only electrode plays the key role in reduction of the electrical resistance with the

tensile strain. Details of the SEM specimen preparation for a stretched electrode are explained in the Experimental Section and shown in Figure S1. However, the exposed composite-only electrode still has high initial resistance, which limits its applications to practical devices and interconnections. To solve this issue, we additionally inkjet-printed silver thin films on top of the exposed composite-only electrodes. Typical silver or gold electrodes keep their low resistance at low tensile strain, but lose their initial properties at higher strain.^[7–12] Therefore, it is expected that two opposite behaviors of the exposed composite-only and silver electrodes would compensate for each other, resulting in an extremely highly stretchable electrode property. Figure 3a shows properties of the inkjet-printed silver electrode on PDMS substrate, the exposed composite-only electrode, (14.3 vol%) and the exposed silver-covered composite electrode. (14.3 vol%) The initial resistance of the inkjet-printed silver, exposed composite-only and silver-covered composite electrodes were 3.2, 237.4 and 2.2 Ω, respectively. The inkjet-printed silver electrode showed a good electrical performance at low tensile strain condition (under 20%). However, its electrical resistance rapidly increased with the tensile strain and it loses electrical conduction at above 38% tensile strain. This increase in resistance originates from the appearance of metal cracks under the tensile strain.^[7–12] The exposed composite-only electrode showed poor electrical performance at low tensile strain condition but its electrical resistance decreased with the tensile strain. The resistance reached 20 Ω at 21% tensile strain and it remained under 20 Ω up to 100% tensile strain. When combined together, the exposed silver-covered composite electrodes showed low resistance at both low and high strain conditions. It showed resistance lower than 20 Ω within range of the experimented tensile strain conditions. Its resistance increased gradually but slightly up to 20% strain, mainly due to crack formation of the inkjet-printed silver thin films. As the tensile strain further increased, the resistance of the exposed silver-covered composite electrodes decreased and became saturated. For visual demonstration, we connected those electrodes to commercial blue LEDs. Their light-emission images at 0, 15, 30, 50 and 100% tensile strain conditions are included in Figure 3b. The operating results are consistent with the measured electrical resistance variation with the tensile strain for three types of electrodes. It is noted that the electrical resistance of the composite-based electrode can be varied if there is any resistive heating. Since temperature dependency of the resistance of the composite-based electrode is common observation,^[20,21] if we want to use our electrodes for high power applications, heating effect should be considered accordingly.

Another promising application of this negative strain-dependency of the exposed composite-only electrodes is implementation of the stretchable display or lighting device with constant image resolution under the tensile strain conditions. It should be noted that most of the previous reports demonstrated stretchable lighting device but with decreasing image resolution with the tensile strain.^[1,22] However, in our approach, we can configure hidden pixels between light-emitting pixels so that the hidden pixels are turned on to maintain the original image resolution when the tensile strain is applied. When the exposed composite-only and exposed silver-covered composite electrodes are appropriately configured, we can produce

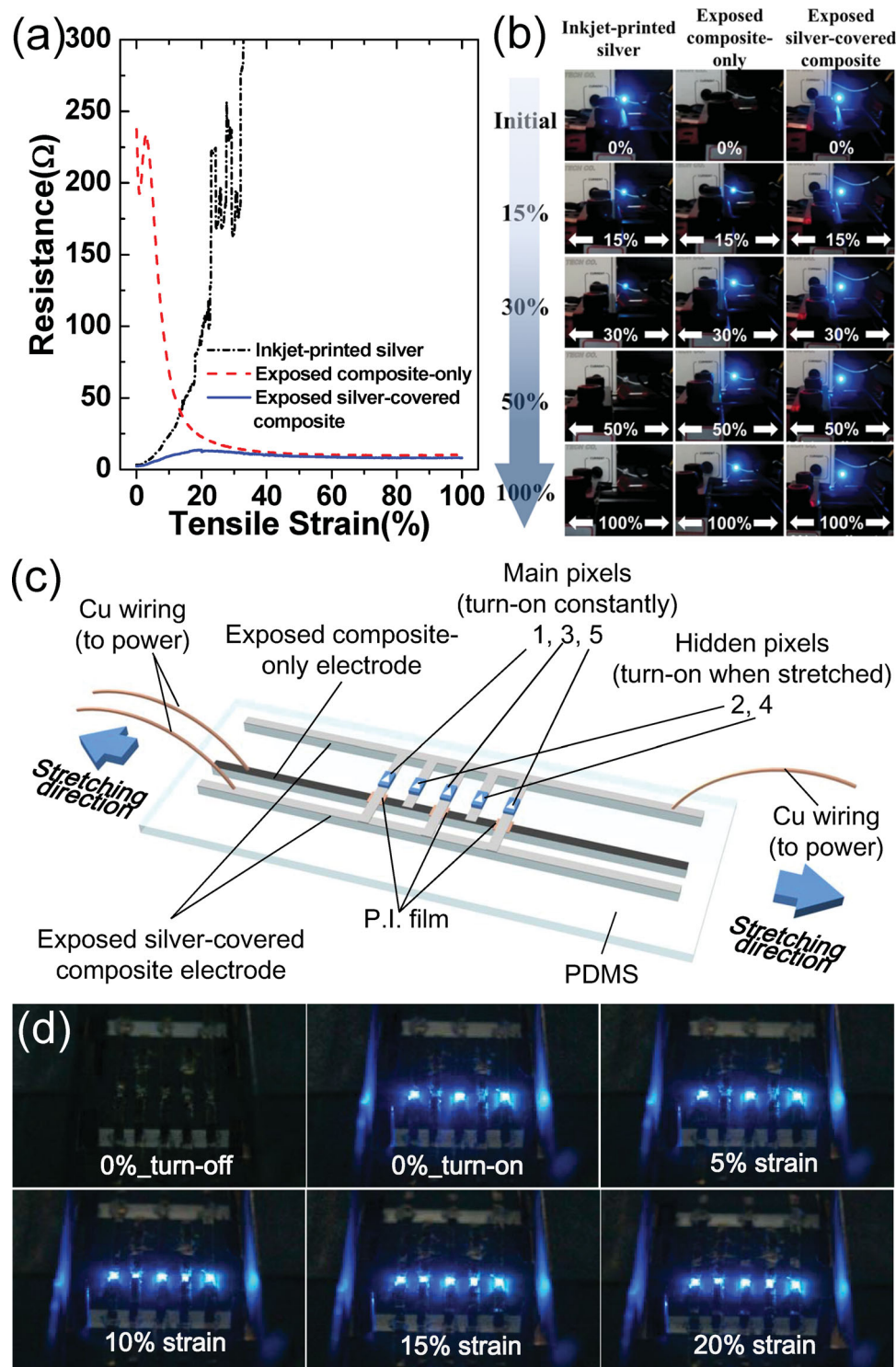


Figure 3. (a) Resistance changes in terms of tensile strain for inkjet-printed silver, exposed composite-only and exposed silver-covered composite. (b) Luminance change of LED which was connected inkjet-printed silver, exposed composite-only and exposed silver-covered composite respectively, with tensile strain. (c) Schematic illustration of resolution-sustainable stretchable lighting device with tensile strain. (d) Luminance change of resolution-sustainable stretchable lighting device with tensile strain.

a resolution-sustainable stretchable lighting device, where number of lighting pixels per unit area can be maintained same under the tensile strain. Figure 3c shows a demonstration

of one-dimensional stretchable lighting device with five surface mount LEDs. Three constantly on (1, 3, 5) and two hidden (2, 4) LEDs are connected to exposed silver-covered composite

and exposed composite-only electrodes, respectively, so that the hidden LEDs are turned on when tensile strain is applied. As the tensile strain increases, the distance between constantly on pixels increases, resulting in reduction of the image resolution. However, at certain tensile strain, the hidden pixels start to be turned on and they will maintain the initial image resolution of the lighting device. Figure 3d shows images taken at various strain conditions up to 20%. The hidden (2, 4) LEDs become turned on with the tensile strain and showed their full brightness at around 15% tensile strain, maintaining it up to 20%. It is noted that resistance of the exposed composite-only electrode becomes similar to that of the exposed silver-covered electrode around 20% tensile strain as shown in Figure 3a. Therefore, we observed similar brightness from the LEDs connected to each electrode. When we optimize the concentration of fillers and improve connection between LEDs themselves with the electrodes, we can control turn-on strain for the hidden LEDs and apply higher tensile strain to our devices.

In conclusion, we have developed a highly stretchable electrode and demonstrated a resolution sustaining lighting device by fully utilizing the unique property of negative strain-dependency in electrical resistance of the magnetically patterned and arranged nickel composite. Resistance of the exposed silver-covered composite electrode was maintained below 20 Ω even up to 100% tensile strain. Although cyclic behavior of the pure nickel composite needs more improvement, nickel-based composite materials have excellent advantages over other materials in terms of simple patterning and in-situ embedding in the matrix. In addition, when nickel-containing composites, such as nickel-combined CNTs^[23] or nickel-attached polymers,^[24] are used, it is expected that more stable, highly-stretchable, patterned interconnecting electrodes can be implemented. Finally, from the conceptual demonstration of the resolution-sustainable lighting device, we believe that our novel technology would be one of the key enabling technologies in implementing future stretchable electronic display devices.

Experimental Section

Fabrication of Exposed Composite-Only Electrodes: Various volume ratio of nickel powder (average particle size $\sim 5 \mu\text{m}$, 99.99% from Sigma Aldrich) was mixed with 10 : 1 weight ratio mixture of poly(dimethylsiloxane) (PDMS; Sylgard 184 from Dow Corning) base and curing agent with a hand blender for 10 minute and then was put in a desiccator for 1 hour to degas air in it. The degassed mixture was cast onto a flat aluminum mold and then placed between two magnetic field modulators and neodymium permanent magnets. (4,000 G) We waited for 10 minute in air at room temperature for the filler patterning and arrangement before it was cured in a convection oven at 130 $^{\circ}\text{C}$ for 30 min.

Fabrication of Exposed Silver-Covered Composite Electrodes: On top of the fully cured exposed composite-only electrodes, silver thin films were inkjet-printed. Nanoparticle type silver ink (DGP-40 from ANP Corp.) was inkjet-printed by using a piezo electric inkjet-printer (DMP-2831 from Dimatix Corp.) after the composite surface was treated with UV ozone cleaner (AH1700 from AHTECH LTS) for 10 minutes to increase the surface energy to be more hydrophilic. Power of the UV light was 28 mWcm^{-2} . Inkjet-printing parameters such as the cartridge temperature, stage temperature, ink drop speed and drop spacing were 30 $^{\circ}\text{C}$, 60 $^{\circ}\text{C}$, 6 ms^{-1} and 25 μm , respectively. The inkjet-printed silver electrode was sintered on a hot plate at 100 $^{\circ}\text{C}$ for 1h.

Fabrication of Stretchable Lighting Device: Three parallel exposed composite-only electrodes embedded in single PDMS matrix were

fabricated using the same method as mentioned above. Then, silver thin films were inkjet-printed onto two outer exposed composite-only electrodes. Anode and cathode of the five surface mount LEDs were attached between the exposed silver-covered composite and composite-only electrodes. Those of the first, third, and fifth LEDs were connected to two exposed silver-covered composite electrodes via inkjet-printed silver. Polyimide films were located on the exposed composite-only electrode in order to prevent unwanted electrical connections. The anode and cathode of the second and fourth LEDs were connected to composite-only electrodes and exposed silver-covered composite, respectively, via inkjet-printed silver.

Characterization: (stretching experiment) Tensile strain was applied using a home-made stretching equipment that can be programmed to control stretching amount and speed. Electrical property was measured using digital sourcemeter (Keithley 2420) that is controlled by a computer. All samples have 10-mm length, 1-mm width, and 1-mm thickness. The tensile strain was applied at 16.7 $\mu\text{m/s}$, and electrical resistance was measured every 0.2 sec. For stretching demonstration, we connected a commercial light emitting diode (LED) with the stretchable electrodes and monitored the light emission under various tensile strain conditions.

Scanning Electron Microscopy: First, exposed composite-only electrode was cut in half (Figure S1a, b, c), along the center line in its stretching direction. Then, specimen was stretched and fixed at a specific tensile strain by using the manually implemented stretching machine (Figure S1d). The stretched specimen which is fixed on the manual stretching machine was carefully and firmly adhered to SEM specimen holder by using conductive carbon tape under the specimen and kapton tape on the side area of the specimen (Figure S1e) in order to maintain the stretched state after removing the stretching machine. As shown in the Figure S1f, the tapes can hold the stretched specimen without the stretching machine. SEM images were taken by field emission SEM. (Model S-4800 from Hitachi)

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

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